

Last login: Tue Jun 2 20:05:29 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % psql -q postgres

postgres=# \c tpch
tpch=# SET ROLE alice;

WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT

```
:stmt
{
  :stmt
  {VARIABLESETSTMT
  :kind 0
  :name role
  :args (
    {A_CONST
    :val "alice"
    :location 9
    }
  )
  :jumble_args false
  :is_local false
  :location 9
  }
  :stmt_location 0
  :stmt_len 14
}
```

WARNING: ===== ANALYZED QUERY =====
DETAIL: {QUERY

```
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{VARIABLESETSTMT
:kind 0
:name role
:args (
  {A_CONST
  :val "alice"
  :location 9
  }
)
:jumble_args false
:is_local false
:location 9
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSublinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:table <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 14
}
```

tpch=> EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT

```
o_orderpriority,
COUNT(*) AS order_count
FROM orders_secure
WHERE o_orderdate >= DATE '1994-01-01'
AND o_orderdate < DATE '1994-01-01' + INTERVAL '3' MONTH
AND EXISTS (
  SELECT 1
  FROM lineitem
  WHERE l_orderkey = o_orderkey
  AND l_commitdate < l_receiptdate
)
```

GROUP BY o_orderpriority
ORDER BY o_orderpriority;

WARNING: ===== RAW PARSE TREE =====
DETAIL: {RAWSTMT

```
:stmt
{EXPLAINSTMT
:query
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
  {RETARGET
  :name <>
  :indirection <>
  :val
  {COLUMNREF
  :fields ("o_orderpriority")
  :location 47
  }
  :location 47
  }
  {RETARGET
  :name order_count
  :indirection <>
  :val
  {FUNCCALL
  :funcname ("count")
  :args <>
  :agg_order <>
  :agg_filter <>
  :over <>
  :agg_within_group false
  :agg_star true
  :agg_distinct false
  :func_variadic false
  :funcformat 0
  :location 68
  }
  :location 68
  }
}
:fromClause (
  {RANGEVAR
  :catalogname <>
  :schemaname <>
  :relname orders_secure
  :inh true
  :relpersistence p
  :alias <>
}
```

```

:location 97
}
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name (">=")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 117
}
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 137
}
:typeName
{TYPENAME
:names ("date")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 132
}
:location -1
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 129
}
{A_EXPR
:name ("<")
:lexpr
{COLUMNREF
:fields ("o_orderdate")
:location 156
}
}
:rexpr
{A_EXPR
:name ("+")
:lexpr
{TYPECAST
:arg
{A_CONST
:val "\1994-01-01"
:location 175
}
:typeName
{TYPENAME
:names ("date")
:typeOid 0
:setof false
:pct_type false
:typmods <>
:typemod -1
:arrayBounds <>
:location 170
}
:location -1
}
}
:rexpr
{TYPECAST
:arg
{A_CONST
:val "\3"
:location 199
}
:typeName
{TYPENAME
:names ("pg_catalog" "interval")
:typeOid 0
:setof false
:pct_type false
:typmods (
{A_CONST
:val 2
:location 203
}
)
:typemod -1
:arrayBounds <>
:location 190
}
:location -1
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 188
}
:rexpr_list_start 0
:rexpr_list_end 0
:location 168
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:opName <>
:subselect
{SELECTSTMT
:distinctClause <>
:intoClause <>
:targetList (
{RESTARTARGET
:name <>
:indirection <>
:val
{A_CONST
:val 1
:location 239
}
}
:location 239
}
)
}
:fromClause (
{RANGEVAR
:catalogname <>
:schemaname <>
:relName lineitem
:inh true
:relPersistence p
:alias <>
:location 254
}
)
)
:whereClause
{BOOLEXP
:boolop and
:args (
{A_EXPR
:name ("=")
:lexpr
{COLUMNREF
:fields ("l_orderkey")
:location 277
}
}
:rexpr
{COLUMNREF
:fields ("o_orderkey")
:location 290
}
}
)
}

```

```

        }
        :rexpr_list_start 0
        :rexpr_list_end 0
        :location 288
    }
    {A_EXPR
    :name ("<")
    :lexpr
    {COLUMNREF
    :fields ("l_commitdate")
    :location 315
    }
    :rexpr
    {COLUMNREF
    :fields ("l_receiptdate")
    :location 330
    }
    :rexpr_list_start 0
    :rexpr_list_end 0
    :location 328
    }
    )
    :location 311
    }
    :groupClause <>
    :groupDistinct false
    :havingClause <>
    :windowClause <>
    :valuesLists <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :lockingClause <>
    :withClause <>
    :op 0
    :all false
    :larg <>
    :rarg <>
    }
    :location 215
    }
    )
    :location 152
    }
    :groupClause (
    {COLUMNREF
    :fields ("o_orderpriority")
    :location 357
    }
    )
    :groupDistinct false
    :havingClause <>
    :windowClause <>
    :valuesLists <>
    :sortClause (
    {SORTBY
    :node
    {COLUMNREF
    :fields ("o_orderpriority")
    :location 382
    }
    :sortby_dir 0
    :sortby_nulls 0
    :useOp <>
    :location -1
    }
    )
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :lockingClause <>
    :withClause <>
    :op 0
    :all false
    :larg <>
    :rarg <>
    }
    :options (
    {DEFELEM
    :defnamespace <>
    :defname analyze
    :arg <>
    :defaction 0
    :location 9
    }
    {DEFELEM
    :defnamespace <>
    :defname verbose
    :arg <>
    :defaction 0
    :location 18
    }
    {DEFELEM
    :defnamespace <>
    :defname buffers
    :arg <>
    :defaction 0
    :location 27
    }
    )
    :stmt_location 0
    :stmt_len 397
    }

```

WARNING: ===== ANALYZED QUERY =====

```

DETAIL: {QUERY
:commandType 6
:querySource 0
:canSetTag true
:utilityStmt
{EXPLAINSTMT
:query
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalpri
ce" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority"
"o_comment")
}
}
:rtekind 0
:relid 16602
:inh true
:relkind v
:rellockmode 1
:perminfoindex 1

```

```

:tablesample <>
:lateral false
:inFromCI true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCI false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16602
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 117
}
)
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 137
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 129
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 156
}
)
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 175
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 199
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 188
}
}
:location 168
}
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
}

```

```

:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_
linenumber" "l_quantity" "l_extendedprice" "l_disco
unt" "l_tax" "l_returnflag" "l_linestatus" "l_shipd
ate" "l_commitdate" "l_receiptdate" "l_shipinstruct
" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh true
:rtekind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 19 20)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 277
}
)
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 290
}
)
:location 288
}
{OEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 315
}
)
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 330
}
)
:location 328
}
)
:location 311
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (

```

```

        {TARGETENTRY
        :expr
        {CONST
        :consttype 23
        :consttypmod -1
        :constcollid 0
        :constlen 4
        :constbyval true
        :constisnull false
        :location 239
        :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :resortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
        }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause <>
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause <>
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
    }
    :location 215
    }
    )
    :location 152
    }
    }
    :mergeActionList <>
    :mergeTargetRelation 0
    :mergeJoinCondition <>
    :targetList (
    {TARGETENTRY
    :expr
    {VAR
    :varno 2
    :varattno 1
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 2
    :varattnosyn 1
    :location -1
    }
    :resno 1
    :resname o_orderpriority
    :resortgroupref 1
    :resorigtbl 16602
    :resorigcol 6
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggsplit 0
    :aggno -1
    :aggtransno -1
    :location 60
    }
    :resno 2
    :resname order_count
    :resortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    )
    :override 0
    :onConflict <>
    :returningOldAlias <>
    :returningNewAlias <>
    :returningList <>
    :groupClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
    )
    :groupDistinct false
    :groupingSets <>
    :havingQual <>
    :windowClause <>
    :distinctClause <>
    :sortClause (
    {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
    }
    )
    :limitOffset <>
    :limitCount <>
    :limitOption 0
    :rowMarks <>
    :setOperations <>
    :constraintDeps <>
    :withCheckOptions <>
    :stmt_location 0
    :stmt_len 0
    }
    :options (
    {DEFELEM

```

```

:dfnamespace <>
:dfname analyze
:arg <>
:dfaction 0
:location 9
}
{DEFELEM
:dfnamespace <>
:dfname verbose
:arg <>
:dfaction 0
:location 18
}
{DEFELEM
:dfnamespace <>
:dfname buffers
:arg <>
:dfaction 0
:location 27
}
}
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable <>
:rteperminfos <>
:jointree <>
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList <>
:overrides 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 397
}

WARNING: ===== REWRITE INPUT =====
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16602
:inh true
:relkind v
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:varatype 25
:varitypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16602
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 0 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1

```

```

    }
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 117
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 137
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 129
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:varatype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 156
}
{OPEXP
:opno 1876
:opfuncid 2071
:opresulttype 1114
:opretset false
:opcollid 0
:inputcollid 0
:args (
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 175
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
{CONST
:consttype 1186
:consttypmod 196607
:constcollid 0
:constlen 16
:constbyval false
:constisnull false
:location 199
:constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
}
)
:location 188
}
)
:location 168
}
{SUBLINK
:sublinkType 0
:sublinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGEBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
}
:rtkind 0
:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2

```

```

:checkAsUser 0
:selectedCols (b 8 19 20)
:insertedCols (b)
:updatedCols (b)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 277
}
)
{VAR
:varno 1
:varattno 1
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 290
}
)
:location 288
}
{OPEXP
:opno 1005
:opfuncid 1007
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 315
}
)
{VAR
:varno 1
:varattno 13
:varstype 1082
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 330
}
)
:location 328
}
)
:location 311
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:conststype 23
:conststypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 239
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:location 215
}
:location 152
}
}
:mergeActionList <>

```

```

:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
      {VAR
        :varno 2
        :varattno 1
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
      }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16602
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
      {AGGREF
        :aggfnoid 2803
        :aggtype 28
        :agccollid 0
        :inputcollid 0
        :aggtranstype 0
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        :agglevelsup 0
        :aggsplit 0
        :aggnosyn -1
        :aggtransno -1
        :location 68
      }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
)

```

WARNING: ===== REWRITE OUTPUT =====

```

DETAIL: (
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
    :utilityStmt <>
    :resultRelation 0
    :hasAggs true
    :hasWindowFuncs false
    :hasTargetSRFs false
    :hasSubLinks true
    :hasDistinctOn false
    :hasRecursive false
    :hasModifyingCTE false
    :hasForUpdate false
    :hasRowSecurity false
    :hasGroupRTE true
    :isReturn false
    :cteList <>
    :rtable (
      {RANGETBENTRY
        :alias <>
        :eref
          {ALIAS
            :aliasname orders_secure
            :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
          }
        :rtekind 1
      }
    )
    :subquery
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (

```

```

{RANGETBLENTTRY
:alias <>
:eref
{ALIAS
:aliasname orders
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalpri
ce" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority"
"o_comment")
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
)
:quals
{SUBLINK
:sublinkType 0
:sublinkid 0
:testexpr <>
:opName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rttable (
{RANGETBLENTTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationk
ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
)
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
)
}
:location -1
}
)

```

```

{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXPR
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 ]
}
)
:location -1
}
)
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
}
{TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname o_orderstatus
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname o_totalprice
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment

```

```

:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 16602
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16602
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 1090
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 117
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 137
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
)
:location 129
}
{OPEXP
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 156
}
{OPEXP
:opno 1076
:opfuncid 2071
:opresulttype 1114
:opretset false

```

```

:opcollid 0
:inputcollid 0
:args (
  {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 175
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
  }
  {CONST
    :consttype 1186
    :consttypmod 196607
    :constcollid 0
    :constlen 16
    :constbyval false
    :constisnull false
    :location 199
    :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
  }
)
:location 188
}
}
:location 168
}
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTRY
    :alias <>
    :eref
    {ALIAS
      :aliasname lineitem
      :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
    }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 19 20)
    :insertedCols (b)
    :updatedCols (b)
  }
)
}
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
    :rtindex 1
  }
)
}
:quals
{BOOLEXP
:boolop and
:args (
  {OPEXP
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 277
      }
      {VAR
        :varno 1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 1
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 1
        :location 290
      }
    )
  }
  :location 288
}
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
  {VAR
    :varno 1
    :varattno 12
    :vartype 1082
  }
)
}

```

```

        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 12
        :location 315
    }
    {VAR
        :varno 1
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location 330
    }
    }
    :location 328
}
}
:location 311
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {CONST
            :consttype 23
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location 239
            :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
        }
        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 215
}
}
:location 152
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
        :expr
        {VAR
            :varno 2
            :varattno 1
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 2
            :varattnosyn 1
            :location -1
        }
        :resno 1
        :resname o_orderpriority
        :ressortgroupref 1
        :resorigtbl 16602
        :resorigcol 6
        :resjunk false
    }
)
{TARGETENTRY
:expr
{AGGREF
    :aggfnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes <>
    :aggdirectargs <>
    :args <>
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvarladic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 0
    :aggno -1
    :aggtransno -1
    :location 68
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
    {SORTGROUPCLAUSE
        :tleSortGroupRef 1
        :eqop 98
    }
)

```

```

:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
    :tleSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
)

```

WARNING: ===== PLANNER INPUT =====

```

DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias <>
    :eref
      {ALIAS
        :aliasname orders_secure
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
      }
    :rtekind 1
    :subquery
      {QUERY
        :commandType 1
        :querySource 0
        :canSetTag true
        :utilityStmt <>
        :resultRelation 0
        :hasAggs false
        :hasWindowFuncs false
        :hasTargetSRFs false
        :hasSubLinks true
        :hasDistinctOn false
        :hasRecursive false
        :hasModifyingCTE false
        :hasForUpdate false
        :hasRowSecurity false
        :hasGroupRTE false
        :isReturn false
        :cteList <>
        :rtable (
          {RANGETBLENTY
            :alias <>
            :eref
              {ALIAS
                :aliasname orders
                :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
              }
            :rtekind 0
            :relid 16561
            :inh true
            :relkind r
            :rellockmode 1
            :permindex 1
            :tablesample <>
            :lateral false
            :inFromCl true
            :securityQuals <>
          }
        )
        :rteperminfos (
          {RTEPERMISSIONINFO
            :relid 16561
            :inh true
            :requiredPerms 2
            :checkAsUser 10
            :selectedCols (b 8 9 10 11 12 13 14 15 16)
            :insertedCols (b)
            :updatedCols (b)
          }
        )
      }
    :jointree
      {FROMEXPR
        :fromlist (
          {RANGETBLREF
            :rtindex 1
          }
        )
        :quals
          {SUBLINK
            :subLinkType 0
            :subLinkId 0
            :testexpr <>
            :opName <>
            :subselect
              {QUERY
                :commandType 1
                :querySource 0
                :canSetTag true
                :utilityStmt <>
                :resultRelation 0
                :hasAggs false
                :hasWindowFuncs false
                :hasTargetSRFs false
                :hasSubLinks false
                :hasDistinctOn false
                :hasRecursive false
                :hasModifyingCTE false
                :hasForUpdate false
                :hasRowSecurity false
                :hasGroupRTE false
                :isReturn false
                :cteList <>
                :rtable (
                  {RANGETBLENTY

```

```

:alias
{
  ALIAS
  :aliasname c
  :colnames <>
}
:eref
{
  ALIAS
  :aliasname c
  :colnames ("c_custkey" "c_name" "c_address" "c_nationk
ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellockmode 1
:permindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
:location -1
}
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]
}
}
:location -1
}
}
:location -1
}
)
:location -1
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetlist (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
}
)

```

```

        :resno 1
        :resname ?column?
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
}
:override 0
:onConflict <>
:ReturningOldAlias <>
:ReturningNewAlias <>
:ReturningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
(TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
(TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname o_custkey
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
(TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname o_orderstatus
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
(TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname o_totalprice
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 4
:resjunk false
}
(TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
(TARGETENTRY

```

```

:expr
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <
:returningOldAlias <
:returningNewAlias <
:returningList <
:groupClause <
:groupDistinct false
:groupingSets <
:havingQual <
>windowClause <
:distinctClause <
:sortClause <
:limitOffset <
:limitCount <
:limitOption 0
:rowMarks <
:setOperations <
:constraintDeps <
:withCheckOptions <
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 16602
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <
}
{RANGEBLENTY
:alias <
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs (
{VAR
:varno 1
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 47
}
)
:lateral false
:inFromCl false
:securityQuals <
}
)

```

```

:rteperminfos (
  {RTEPERMISSIONINFO
  :releid 16602
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 0 12 13)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
    :rtindex 1
    }
  )
:quals
  {BOOLEXP
  :boolop and
  :args (
    {OPEXP
    :opno 1098
    :opfuncid 1090
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
      :varno 1
      :varattno 5
      :vartype 1082
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 5
      :location 117
      }
      {CONST
      :consttype 1082
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location 137
      :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
      }
    )
    :location 129
  )
  {OPEXP
  :opno 2345
  :opfuncid 2338
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 0
  :args (
    {VAR
    :varno 1
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 156
    }
    {OPEXP
    :opno 1076
    :opfuncid 2071
    :opresulttype 1114
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {CONST
      :consttype 1082
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location 175
      :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
      }
      {CONST
      :consttype 1186
      :consttypmod 196607
      :constcollid 0
      :constlen 16
      :constbyval false
      :constisnull false
      :location 199
      :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
      }
    )
    :location 188
  )
  }
  :location 168
}
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:opetName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rttable (
  {RANGETBLENTRY
  :alias <>
  :eref
  {ALIAS
  :aliasname lineitem
  :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
  }
  :rtekind 0
}

```

```

:relid 16566
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 19 20)
:insertedCols (b)
:updatedCols (b)
}
)
:jointree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 277
}
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 290
}
)
:location 288
}
{OPEXP
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 315
}
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 330
}
)
:location 328
}
)
:location 311
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 239
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:resortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>

```

```

        :limitCount <>
        :limitOption 0
        :rowMarks <>
        :setOperations <>
        :constraintDepts <>
        :withCheckOptions <>
        :stmt_location 0
        :stmt_len 0
    }
    :location 215
}
)
:location 152
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16002
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 68
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDepts <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

```

WARNING: ===== FINAL PLAN =====

```

DETAIL: {AGG
:plan.disabled_nodes 0
:plan.startup_cost 194594.51589866364
:plan.total_cost 194617.5859964136
:plan.plan_rows 5
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe false
:plan.async_capable false
:plan.plan_node_id 0
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16002
:resorigcol 6
:resjunk false
}
}

```

```

{TARGETENTRY
:expr
  {AGGREF
    :aggnoid 2803
    :aggtype 20
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 20
    :aggargtypes <>
    :aggdirectargs <>
    :args (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 2
            :vartype 20
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
          :resno 1
          :resname <>
          :ressortgroupref 0
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
    :aggorder <>
    :aggdistinct <>
    :aggfilter <>
    :aggstar true
    :aggvariadic false
    :aggkind n
    :aggpresorted false
    :agglevelsup 0
    :aggssplit 9
    :aggno 0
    :aggtransno 0
    :location 68
  }
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
  {GATHERMERGE
    :plan.disabled_nodes 0
    :plan.startup_cost 194594.51589866364
    :plan.total_cost 194617.47599641362
    :plan.plan_rows 12
    :plan.plan_width 17
    :plan.parallel_aware false
    :plan.parallel_safe false
    :plan.async_capable false
    :plan.plan_node_id 1
    :plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 1
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
          :resno 1
          :resname <>
          :ressortgroupref 1
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
    :plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 2
            :vartype 20
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
          :resno 2
          :resname <>
          :ressortgroupref 0
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
    }
  )
:plan.qual <>
:plan.lefttree
  {AGG
    :plan.disabled_nodes 0
    :plan.startup_cost 193594.49212422612
    :plan.total_cost 193616.06712422607
    :plan.plan_rows 5
    :plan.plan_width 17
    :plan.parallel_aware false
    :plan.parallel_safe true
    :plan.async_capable false
    :plan.plan_node_id 2
    :plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 1
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
          :resno 1
          :resname <>
          :ressortgroupref 1
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
    }
  )

```

```

(TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 20
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 6
:aggno 0
:aggtransno 0
:location 68
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SORT
:plan.disabled_nodes 0
:plan.startup_cost 193594.49212422612
:plan.total_cost 193401.6671242261
:plan.plan_rows 2870
:plan.plan_width 9
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 3
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:plan.qual <>
:plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 154455.46666666667
:join.plan.total_cost 193429.6560416667
:join.plan.plan_rows 2870
:join.plan.plan_width 9
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 4
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:join.plan.qual <>
:join.plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 4806.4125
:join.plan.total_cost 284374999996
:join.plan.plan_rows 10210
:join.plan.plan_width 13
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 5
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)

```

```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 1
        :location -1
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{SEQSCAN
    :scan.plan.disabled_nodes 0
    :scan.plan.startup_cost 0
    :scan.plan.total_cost 35383
    :scan.plan.plan_rows 22585
    :scan.plan.plan_width 17
    :scan.plan.parallel_aware true
    :scan.plan.parallel_safe true
    :scan.plan.async_capable false
    :scan.plan.plan_node_id 6
    :scan.plan.targetlist (
        {TARGETENTRY
            :expr
            {VAR
                :varno 4
                :varattno 1
                :vartype 23
                :vartypmod -1
                :varcollid 0
                :varnullingrels (b)
                :varlevelsup 0
                :varreturningtype 0
                :varnosyn 4
                :varattnosyn 1
                :location -1
            }
        }
        :resno 1
        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
    {TARGETENTRY
        :expr
        {VAR
            :varno 4
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 4
            :varattnosyn 2
            :location -1
        }
    }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
}
{TARGETENTRY
    :expr
    {VAR
        :varno 4
        :varattno 3
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 3
        :location -1
    }
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
    :expr
    {VAR
        :varno 4
        :varattno 4
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 4
        :location -1
    }
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
    :expr
    {VAR
        :varno 4
        :varattno 5
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 5
        :location -1
    }
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
    :expr
    {VAR
        :varno 4
        :varattno 6
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
    }
}

```

```

        :varnosyn 4
        :varattnosyn 6
        :location -1
    }
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 7
:location -1
}
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 8
:location -1
}
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 9
:location -1
}
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 137
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
}
:location 129
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
}
:location 168
}
)

```

```

:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 4
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 4453.25
:plan.total_cost 4453.25
:plan.plan_rows 28253
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 7
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 4453.25
:scan.plan.plan_rows 28253
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 8
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
)
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 5
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
)
:skewTable 16561
:skewColumn 2
:skewInherit false
:rows_total 67807.2
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)

```

```

:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno -1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 1
        :location -1
      }
    )
  }
  :location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location -1
  }
)
}
:join.plan.righttree
{HASH
  :plan.disabled_nodes 0
  :plan.startup_cost 135974.79166666666
  :plan.total_cost 135974.79166666666
  :plan.plan_rows 833461
  :plan.plan_width 4
  :plan.parallel_aware true
  :plan.parallel_safe true
  :plan.async_capable false
  :plan.plan_node_id 9
  :plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno -2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location -1
      }
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
  :scan.plan.disabled_nodes 0
  :scan.plan.startup_cost 0
  :scan.plan.total_cost 135974.79166666666
  :scan.plan.plan_rows 833461
  :scan.plan.plan_width 4
  :scan.plan.parallel_aware true
  :scan.plan.parallel_safe true
  :scan.plan.async_capable false
  :scan.plan.plan_node_id 10
  :scan.plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno 3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location 277
      }
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:scan.plan.qual (
  {OPEXPR
    :opno 1095
    :opfuncid 1087
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
      }
    )
  }
)

```

```

:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 315
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 330
}
}
:location 328
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 277
}
)
:skewTable 16561
:skewColumn 1
:skewInherit false
:rows_total 2000306.4
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
{VAR
:varno -1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 277
}
)
}
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:aggstrategy 1
:aggssplit 6
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:num_workers 2
:rescan_param 0

```

```
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
:initParam (b)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 1
:aggssplit 9
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
```

QUERY PLAN

```
-----
Finalize GroupAggregate (cost=194594.52..194617.59 rows=5 width=17) (actual time=1472.182..1494.702 rows=5.00 loops=1)
  Output: orders.o_orderpriority, count(*)
  Group Key: orders.o_orderpriority
  Buffers: shared hit=4628 read=129788, temp read=20134 written=20552
  -> Gather Merge (cost=194594.52..194617.48 rows=12 width=17) (actual time=1472.021..1494.688 rows=15.00 loops=1)
    Output: orders.o_orderpriority, (PARTIAL count(*))
    Workers Planned: 2
    Workers Launched: 2
    Buffers: shared hit=4628 read=129788, temp read=20134 written=20552
    -> Partial GroupAggregate (cost=193594.49..193616.07 rows=5 width=17) (actual time=1466.190..1466.852 rows=5.00 loops=3)
      Output: orders.o_orderpriority, PARTIAL count(*)
      Group Key: orders.o_orderpriority
      Buffers: shared hit=4628 read=129788, temp read=20134 written=20552
      Worker 0: actual time=1463.738..1464.386 rows=5.00 loops=1
      Buffers: shared hit=1540 read=43176, temp read=6796 written=6900
      Worker 1: actual time=1463.081..1464.477 rows=5.00 loops=1
      Buffers: shared hit=1549 read=43074, temp read=6814 written=6872
      -> Sort (cost=193594.49..193601.67 rows=2870 width=9) (actual time=1466.017..1466.335 rows=7772.33 loops=3)
        Output: orders.o_orderpriority
        Sort Key: orders.o_orderpriority
        Sort Method: quicksort Memory: 193kB
        Buffers: shared hit=4628 read=129788, temp read=20134 written=20552
        Worker 0: actual time=1463.557..1463.874 rows=7962.00 loops=1
          Sort Method: quicksort Memory: 193kB
          Buffers: shared hit=1540 read=43176, temp read=6796 written=6900
        Worker 1: actual time=1463.616..1463.947 rows=8284.00 loops=1
          Sort Method: quicksort Memory: 385kB
          Buffers: shared hit=1549 read=43074, temp read=6814 written=6872
        -> Parallel Hash Semi Join (cost=154455.47..193429.66 rows=2870 width=9) (actual time=1330.852..1463.568 rows=7772.33 loops=3)
          Output: orders.o_orderpriority
          Hash Cond: (orders.o_orderkey = lineitem.l_orderkey)
          Buffers: shared hit=4612 read=129788, temp read=20134 written=20552
          Worker 0: actual time=1334.138..1461.163 rows=7962.00 loops=1
            Buffers: shared hit=1532 read=43176, temp read=6796 written=6900
          Worker 1: actual time=1318.712..1461.301 rows=8284.00 loops=1
            Buffers: shared hit=1541 read=43074, temp read=6814 written=6872
          -> Parallel Hash Semi Join (cost=4806.41..40362.28 rows=10210 width=13) (actual time=20.894..170.633 rows=8461.00 loops=3)
            Output: orders.o_orderpriority, orders.o_orderkey
            Hash Cond: (orders.o_custkey = c.c_custkey)
            Buffers: shared hit=4142 read=25538
            Worker 0: actual time=20.804..170.619 rows=8360.00 loops=1
              Buffers: shared hit=1372 read=8484
            Worker 1: actual time=21.083..170.698 rows=8545.00 loops=1
              Buffers: shared hit=1389 read=8494
            -> Parallel Seq Scan on public.orders (cost=0.00..35383.00 rows=22585 width=17) (actual time=0.755..142.466 rows=18783.00 loops=3)
              Output: orders.o_orderkey, orders.o_custkey, orders.o_orderstatus, orders.o_totalprice, orders.o_orderdate, orders.o_orderpriority, orders.o_clerk, orders.o_shippr
              iority, orders.o_comment
              Filter: ((orders.o_orderdate >= '1994-01-01':date) AND (orders.o_orderdate < '1994-04-01 00:00:00':timestamp without time zone))
              Rows Removed by Filter: 481217
              Buffers: shared hit=470 read=25538
              Worker 0: actual time=0.739..142.475 rows=18726.00 loops=1
                Buffers: shared hit=150 read=8484
              Worker 1: actual time=0.808..142.531 rows=18645.00 loops=1
                Buffers: shared hit=164 read=8494
              -> Parallel Hash (cost=4453.25..4453.25 rows=28253 width=4) (actual time=19.922..19.922 rows=22663.00 loops=3)
                Output: c.c_custkey
                Buckets: 131072 Batches: 1 Memory Usage: 3712kB
                Buffers: shared hit=3672
                Worker 0: actual time=19.732..19.733 rows=22664.00 loops=1
                  Buffers: shared hit=1222
                Worker 1: actual time=20.011..20.011 rows=22671.00 loops=1
                  Buffers: shared hit=1225
                -> Parallel Seq Scan on public.customer c (cost=0.00..4453.25 rows=28253 width=4) (actual time=0.031..15.218 rows=22663.00 loops=3)
                  Output: c.c_custkey
                  Filter: (c.c_acctbal > '5000':double precision)
                  Rows Removed by Filter: 27337
                  Buffers: shared hit=3672
                  Worker 0: actual time=0.035..15.278 rows=22664.00 loops=1
                    Buffers: shared hit=1222
                  Worker 1: actual time=0.022..15.214 rows=22671.00 loops=1
                    Buffers: shared hit=1225
                -> Parallel Hash (cost=135974.79..135974.79 rows=833461 width=4) (actual time=1135.342..1135.342 rows=1264432.00 loops=3)
                  Output: lineitem.l_orderkey
                  Buckets: 262144 (originally 262144) Batches: 32 (originally 16) Memory Usage: 6752kB
                  Buffers: shared hit=470 read=104250, temp read=9125 written=20160
                  Worker 0: actual time=1133.282..1133.283 rows=1262530.00 loops=1
                    Buffers: shared hit=160 read=34692, temp read=3099 written=6772
                  Worker 1: actual time=1133.313..1133.313 rows=1258432.00 loops=1
                    Buffers: shared hit=152 read=34580, temp read=3070 written=6744
                  -> Parallel Seq Scan on public.lineitem (cost=0.00..135974.79 rows=833461 width=4) (actual time=0.524..610.725 rows=1264432.00 loops=3)
                    Output: lineitem.l_orderkey
                    Filter: (lineitem.l_commitdate < lineitem.l_receiptdate)
                    Rows Removed by Filter: 735973
                    Buffers: shared hit=470 read=104250
                    Worker 0: actual time=0.498..614.480 rows=1262530.00 loops=1
                      Buffers: shared hit=160 read=34692
                    Worker 1: actual time=0.378..611.863 rows=1258432.00 loops=1
                      Buffers: shared hit=152 read=34580

Planning:
  Buffers: shared hit=232
  Planning Time: 6.941 ms
  Execution Time: 1495.683 ms
(91 rows)

tpch=>
```